

The multilevel summation method for fast evaluation of electrostatic interactions

A QUICK-START INTRODUCTION

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This is a polished and extended version of slides shown in an internal group presentation on June 4, 2021.

It is meant as a quick-start introduction to the multilevel summation method algorithm, and in particular the flavor thereof we plan to implement, based primarily on J. Chem. Phys. 144, 114112 (2016).

Contents

Introduction

Fundamental ideas

More algorithm details

Schematic cost analysis

Boundary conditions

Introduction

Introduction

Electrostatic energy of a collection of N point charges q_i at positions $\mathbf{r}_i$:

$$U = \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} \frac{q_i q_j}{|\mathbf{r}_i - \mathbf{r}_j|} = \frac{1}{2} (\underline{q})^T \underline{\underline{K}} \underline{q},$$

with

- $(\underline{q})_i = (q_1, \dots, q_N)^T$,
- $(\underline{\underline{K}})_{i,j} = \begin{cases} 1/|\mathbf{r}_i - \mathbf{r}_j|, & i \neq j, \\ 0, & i = j. \end{cases}$

$\frac{1}{2}N(N-1)$ unique values of interaction kernel $\underline{\underline{K}} \implies \mathcal{O}(N^2)$ scaling.

Notable applications:

- Ionic crystals and liquids
- Biomolecular systems
- Equivalent problems appear in other contexts, e.g. gravity

Fast electrostatics methods developed to improve $\mathcal{O}(N^2)$ scaling:

- Particle-Mesh Ewald (PME), Particle-Particle-Particle Mesh (P³M)
- Fast Multipole Method (FMM)
- Multilevel Summation Method (MSM)

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Multilevel Summation Method

- Novel in comparison, relatively less well-known than other electrostatic methods
- Selling points:
 - $\mathcal{O}(N)$ scaling.
 - Supports mixed boundary conditions (periodic along some directions only).
 - Conceptually fairly simple.
 - Permits straightforward application to systems of any dimension (including > 3).
- Much of the more recent development due to David Hardy *et al.*, driven by application in biomolecular systems

- Thesis of David Hardy:

Multilevel Summation for the Fast Evaluation of Forces for the Simulation of Biomolecules., David J. Hardy. Ph.D. thesis, University of Illinois at Urbana-Champaign, 2006.

- Our implementation:

Multilevel summation with B-spline interpolation for pairwise interactions in molecular dynamics simulations., Hardy *et al.*, J. Chem. Phys. 144, 114112 (2016). 3/32

Fundamental ideas

Fundamental ideas

1. Split Coulomb interaction kernel into a sum of functions that are
 - increasingly long-range,
 - increasingly smooth.
2. Approximate these partial kernels by interpolating from increasingly coarser grids.

Fundamental ideas: 1. Kernel splitting

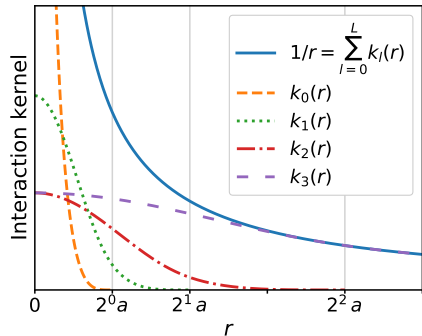
The Coulomb kernel $1/r$ is split into $L + 1$ summands:

$$\frac{1}{r} = \sum_{l=0}^L k_l(r) = k_0(r) + k_1(r) + \cdots + k_{L-1}(r) + k_L(r)$$

Each kernel function

- has a larger range ...
- is more slowly varying with r ...

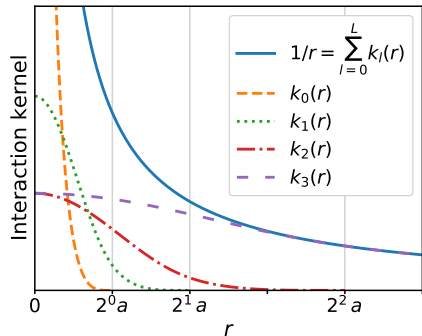
... than the previous one.



Fundamental ideas: 1. Kernel splitting

$$\frac{1}{r} = \sum_{l=0}^L k_l(r) = k_0(r) + k_1(r) + \cdots + k_{L-1}(r) + k_L(r)$$

kernel	cutoff radius	value at $r = 0$
k_0	a	∞
k_1	$2^1 a$	finite
k_2	$2^2 a$	finite
...	...	...
k_{L-1}	$2^{L-1} a$	finite
k_L	∞	finite



Fundamental ideas: 1. Kernel splitting

With the splitting $1/r = \sum_{l=0}^L k_l(r)$, the electrostatic energy is written as

$$\begin{aligned} U &= \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} q_i q_j \frac{1}{r_{ij}} \\ &= \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} q_i q_j k_0(r_{ij}) + \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j \neq i} q_i q_j k_l(r_{ij}) \end{aligned}$$

It will prove useful to avoid the restriction $j \neq i$ in the second term:

$$U = \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} q_i q_j k_0(r_{ij}) - \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N q_i^2 k_l(r)|_{r=0} + \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N q_i q_j k_l(r_{ij})$$

... remember that the $k_l(r)$, $l \geq 1$, have no singularity at zero

Fundamental ideas: 1. Kernel splitting

So far, only rewriting:

$$U = \underbrace{\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} q_i q_j k_0(r_{ij}) - \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N q_i^2 k_l(r)|_{r=0}}_{:=U^0} + \underbrace{\frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N q_i q_j k_l(r_{ij})}_{:=U^{1+}}$$

Next step:

Approximate U^{1+} by interpolating the interaction kernels $k_l(r)$, $l \geq 1$, from grids of increasing coarseness.

Fundamental ideas: 2. Interpolation from grids

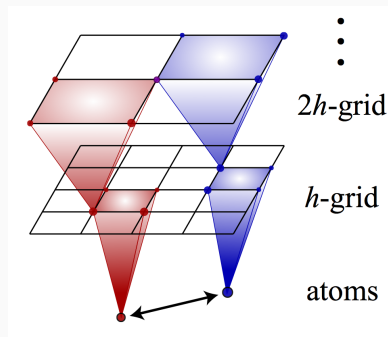
l	grid is defined?	grid spacing
0	×	n.a.
1	✓	h
2	✓	$2h$
...	...	...
$L-1$	✓	$2^{L-2}h$
L	✓	$2^{L-1}h$

Note:

- At level $l = 0$, there is no grid!
- Interaction cutoff radius at level l :
Grid spacing at level l :

$$a_l = 2^l a$$

$$h_l = 2^{l-1} h$$



from <https://www.ks.uiuc.edu/Research/msm/>

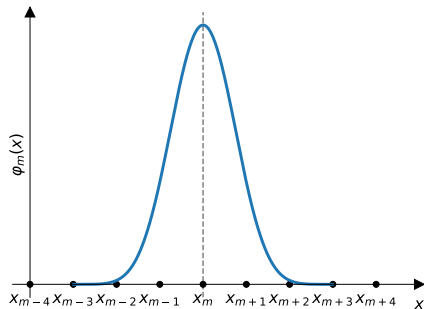
Fundamental ideas: 2. Interpolation from grids

Preliminary: Interpolating a scalar-valued function f

- of a one-dimensional argument x :

$$f(x) \approx \sum_m \hat{f}_m \varphi_m(x), \quad \text{with}$$

- $\hat{f}_m$: coefficients,
- $\varphi_m(x)$: the basis function centered at grid point $x_m = mh$, evaluated at x .



What basis functions might look like.

- of a three-dimensional argument $\mathbf{r} = (x, y, z)^T$:

$$f(\mathbf{r}) \approx \sum_{\mathbf{m}} \hat{f}_{\mathbf{m}} \varphi_{\mathbf{m}}(\mathbf{r}), \quad \text{with } \mathbf{m} = (m_x, m_y, m_z)^T, \varphi_{\mathbf{m}}(\mathbf{r}) = \varphi_{m_x}(x) \varphi_{m_y}(y) \varphi_{m_z}(z).$$

Fundamental ideas: 2. Interpolation from grids

Actual MSM application slightly more complicated:

Need to interpolate the kernels $k_l(|\mathbf{r} - \mathbf{r}'|)$, $l \geq 1$, as a function of *two* positions $\mathbf{r}$, $\mathbf{r}'$,

$$k_l(|\mathbf{r} - \mathbf{r}'|) \approx \sum_{\mathbf{m}} \sum_{\mathbf{n}} \varphi_{\mathbf{m}}^l(\mathbf{r}) \mathcal{K}_{\mathbf{m},\mathbf{n}}^l \varphi_{\mathbf{n}}^l(\mathbf{r}').$$

Since $k_l(|\mathbf{r} - \mathbf{r}'|)$ depends only on the difference of $\mathbf{r}$ and $\mathbf{r}'$, one can simplify

$$\mathcal{K}_{\mathbf{m},\mathbf{n}}^l = (\hat{k}_l)_{\mathbf{m}-\mathbf{n}},$$

with $\hat{k}_l$ as in

$$k_l(|\mathbf{r}|) \approx \sum_{\mathbf{m}} (\hat{k}_l)_{\mathbf{m}} \varphi_{\mathbf{m}}^l(\mathbf{r}).$$

$\implies \mathcal{K}_{\mathbf{m},\mathbf{n}}^l$ can be stored as a three-, rather than six-dimensional, array.

Fundamental ideas: Putting everything together

$$U^{1+} = \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N q_i q_j k_l (|\mathbf{r}_i - \mathbf{r}_j|)$$

$$\approx \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N q_i q_j \left[\sum_{\mathbf{m}} \sum_{\mathbf{n}} \varphi_{\mathbf{m}}^l(\mathbf{r}_i) \mathcal{K}_{\mathbf{m},\mathbf{n}}^l \varphi_{\mathbf{n}}^l(\mathbf{r}_j) \right]$$

$$= \frac{1}{2} \sum_{l=1}^L \sum_{\mathbf{m}} \sum_{\mathbf{n}} \underbrace{\sum_{i=1}^N q_i \varphi_{\mathbf{m}}^l(\mathbf{r}_i)}_{:= q_{\mathbf{m}}^l \text{ ("grid charge")}} \mathcal{K}_{\mathbf{m},\mathbf{n}}^l \underbrace{\sum_{j=1}^N q_j \varphi_{\mathbf{n}}^l(\mathbf{r}_j)}_{:= q_{\mathbf{n}}^l \text{ ("grid charge")}}$$

$$= \frac{1}{2} \sum_{l=1}^L \sum_{\mathbf{m}} \sum_{\mathbf{n}} q_{\mathbf{m}}^l \mathcal{K}_{\mathbf{m},\mathbf{n}}^l q_{\mathbf{n}}^l = \frac{1}{2} \sum_{l=1}^L (\underline{q}^l)^T \underline{\mathcal{K}}^l \underline{q}^l$$

$\sum$: sum over particles

$\sum$: sum over different grid levels

$\sum$: sum over grid points at a given level

Fundamental ideas: Putting everything together

$$\begin{aligned}
 U^{1+} &= \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N q_i q_j k_l(|\mathbf{r}_i - \mathbf{r}_j|) && \leftarrow \text{scales as } \mathcal{O}(N^2) \\
 &\approx \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N q_i q_j \left[\sum_{\mathbf{m}} \sum_{\mathbf{n}} \varphi_{\mathbf{m}}^l(\mathbf{r}_i) \mathcal{K}_{\mathbf{m},\mathbf{n}}^l \varphi_{\mathbf{n}}^l(\mathbf{r}_j) \right] \\
 &= \frac{1}{2} \sum_{l=1}^L \sum_{\mathbf{m}} \sum_{\mathbf{n}} \underbrace{\sum_{i=1}^N q_i \varphi_{\mathbf{m}}^l(\mathbf{r}_i)}_{:= q_{\mathbf{m}}^l \text{ ("grid charge")}} \mathcal{K}_{\mathbf{m},\mathbf{n}}^l \underbrace{\sum_{j=1}^N q_j \varphi_{\mathbf{n}}^l(\mathbf{r}_j)}_{:= q_{\mathbf{n}}^l \text{ ("grid charge")}} && \leftarrow \text{grid charge computations} \\
 & && \text{scale as } \mathcal{O}(N) \\
 &= \frac{1}{2} \sum_{l=1}^L \sum_{\mathbf{m}} \sum_{\mathbf{n}} q_{\mathbf{m}}^l \mathcal{K}_{\mathbf{m},\mathbf{n}}^l q_{\mathbf{n}}^l = \frac{1}{2} \sum_{l=1}^L (\underline{q}^l)^{\top} \underline{\mathcal{K}}^l \underline{q}^l
 \end{aligned}$$

More algorithm details

More algorithm details: Functional form of partial kernels

The partial kernels $k_l(r)$ from $1/r = \sum_{l=1}^L k_l(r)$ are constructed in terms of a dimensionless “softening function” $\sigma(\rho)$,

- $\sigma(\rho) = 1/\rho$ for $\rho \geq 1$,
- higher derivatives bounded for $\rho < 1$.

Dimensionless auxiliary functions:

$$\begin{aligned}\gamma_{\text{short}}(\rho) &= 1/\rho - \sigma(\rho) \\ \gamma_{\text{interm}}(\rho) &= 2\sigma(2\rho) - \sigma(\rho) \\ \gamma_{\text{long}}(\rho) &= 2\sigma(2\rho)\end{aligned}$$

Partial kernels $k_l(r)$:

$$k_l(r) = \begin{cases} \frac{1}{a} \gamma_{\text{short}}\left(\frac{r}{a}\right), & l = 0 \\ \frac{1}{2^l a} \gamma_{\text{interm}}\left(\frac{r}{2^l a}\right), & l = 1, \dots, L-1 \\ \frac{1}{2^L a} \gamma_{\text{long}}\left(\frac{r}{2^L a}\right), & l = L \end{cases}$$

→ Choice of σ and L completely determines the functional form of all the $k_l(r)$.

More algorithm details: Functional form of partial kernels

The article defines the “softening function” σ as

$$\sigma_p(\rho) = \begin{cases} p\text{-th order Taylor expansion of } 1/\rho \text{ around } \rho = 1, & 0 \leq \rho \leq 1 \\ 1/\rho, & \rho > 1 \end{cases},$$

based on a theoretical argument about interpolation accuracy.

Note:

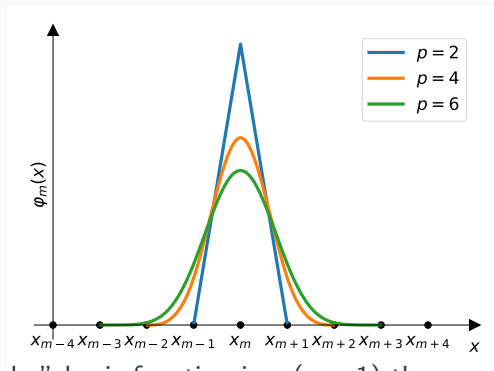
- Polynomial order p of the softening function is a parameter of the algorithm that must be chosen.
- The p -th order softening function σ_p is supposed to be used in conjunction with p -th order interpolation basis functions $\varphi(r)$ (more on that soon).

More algorithm details: Basis functions for interpolation

So far, we haven't specified the $\varphi_{\mathbf{m}}(\mathbf{r}) \rightarrow$ use *B-splines* Φ

$$\varphi_m(x) = \Phi(x/h - m), \quad \begin{cases} \Phi & \text{centered, unscaled B-spline,} \\ h & \text{grid spacing,} \\ m & \text{grid point index.} \end{cases}$$

- Short for “basis splines” (of the space of piecewise polynomials).
- Minimal support.
- Come in different polynomial degrees.



Note: In the language of the article, a “ p -th order” basis function is a $(p - 1)$ -th degree polynomial.

More algorithm details: Interpolation coefficients

$$f(x) \approx \sum_m \hat{f}_m \varphi_m(x) =: \tilde{f}(x)$$

- Computing the interpolant $\tilde{f}(x)$ requires the coefficients $\hat{f}_m$.
- By definition, interpolant $\tilde{f}(x)$ must match target function at grid points $x = mh$,

$$\tilde{f}(mh) \stackrel{!}{=} f(mh).$$

- Yields a system of equations for the coefficients $\hat{f}_m$.

→ But a more convenient way of obtaining the interpolant exists!

More algorithm details: Interpolation coefficients

- Consider instead the *fundamental spline* $\Psi(u)$ defined by the property

$$\Psi(u) = \begin{cases} 1, & u = 0 \\ 0, & u = \pm 1, \pm 2, \dots \end{cases}.$$

- Allows the interpolant to be written as

$$\tilde{f}(x) = \sum_m f(mh) \Psi(x/h - m).$$

- Proof: $\tilde{f}(\textcolor{red}{mh}) = \sum_n f(nh) \Psi(\textcolor{red}{mh}/h - n) = \sum_n f(nh) \underbrace{\Psi(m - n)}_{\delta_{mn}} = f(mh)$

$$\tilde{f}(mh) = f(mh) \checkmark$$

More algorithm details: Interpolation coefficients

- Suppose the fundamental spline can be written as a linear combination

$$\psi(u) = \sum_m \omega_m \Phi(u - m),$$

with Φ the centered, unscaled B-spline as defined previously.

- Solve a system of equations for the fundamental spline coefficients ω_m *just once*.
- With ω_m known, interpolation coefficients $\hat{f}_m$ for *any* target function $f(x)$ can be easily computed as the convolution

$$\hat{f}_m = \sum_n \omega_{m-n} f(nh) \quad (\text{without proof}).$$

More algorithm details: Spline nesting

- B-splines at a higher grid level can be expressed as a linear combination of those at the grid level below:

$$\varphi_m^l(x) = \sum_n (\mathcal{I}_{l-1}^l)_{m,n} \varphi_n^{l-1}(x)$$

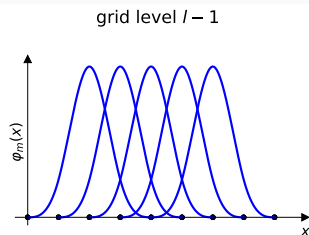
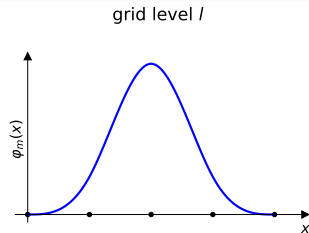
- This permits computing grid charges for $l > 1$ from those at lower levels:

$$\underline{q}^l = \underline{\mathcal{I}}_{l-1}^l \underline{q}^{l-1}$$

- For example:

$$\underline{q}^2 = \underline{\mathcal{I}}_1^2 \underline{q}^1,$$

$$\underline{q}^3 = \underline{\mathcal{I}}_2^3 \underline{\mathcal{I}}_1^2 \underline{q}^1$$



More algorithm details: Spline nesting

With the B-spline nesting property, one can rewrite the grids' energy contribution U^{1+} .

Example for $L = 3$:

$$\begin{aligned} U^{1+} &\approx \frac{1}{2} \sum_{l=1}^3 (\underline{q}^l)^\top \underline{\mathcal{K}}^l \underline{q}^l \\ &= \frac{1}{2} (\underline{q}^1)^\top \underline{\mathcal{K}}^1 \underline{q}^1 + (\underline{q}^2)^\top \underline{\mathcal{K}}^2 \underline{q}^2 + (\underline{q}^3)^\top \underline{\mathcal{K}}^3 \underline{q}^3 \\ &= \frac{1}{2} (\underline{q}^1)^\top \underline{\mathcal{K}}^1 \underline{q}^1 + (\underline{\mathcal{I}}_1^2 \underline{q}^1)^\top \underline{\mathcal{K}}^2 \underline{\mathcal{I}}_1^2 \underline{q}^1 + (\underline{\mathcal{I}}_2^3 \underline{\mathcal{I}}_1^2 \underline{q}^1)^\top \underline{\mathcal{K}}^3 \underline{\mathcal{I}}_2^3 \underline{\mathcal{I}}_1^2 \underline{q}^1 \\ &= \frac{1}{2} (\underline{q}^1)^\top \left[\underline{\mathcal{K}}^1 + \underline{\mathcal{I}}_2^1 \left(\underline{\mathcal{K}}^2 + \underline{\mathcal{I}}_3^2 \underline{\mathcal{K}}^3 \underline{\mathcal{I}}_2^3 \right) \underline{\mathcal{I}}_1^2 \right] \underline{q}^1, \end{aligned}$$

where the notation $\underline{\mathcal{I}}_l^{l-1} := (\underline{\mathcal{I}}_{l-1}^l)^\top$ was used.

More algorithm details: “Ladder diagram”

Example for $L = 3$:

$$U^{1+} \approx \frac{1}{2}(\underline{q}^1)^\top \left[\underline{\kappa}^1 + \underline{\mathcal{I}}_2^1 \left(\underline{\kappa}^2 + \underline{\mathcal{I}}_3^2 \underline{\kappa}^3 \underline{\mathcal{I}}_2^3 \right) \underline{\mathcal{I}}_1^2 \right] \underline{q}^1$$

$$\underbrace{\hspace{15em}}_{\underline{q}^2}$$

$$\underbrace{\hspace{10em}}_{\underline{q}^3}$$

$$\underbrace{\hspace{5em}}_{\underline{\kappa}^3 \underline{q}^3 := \underline{e}^{3+}}$$

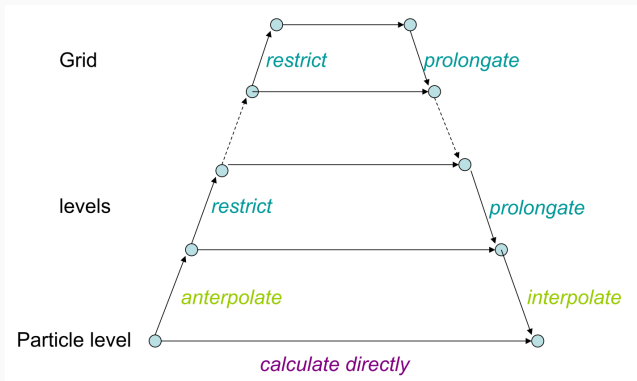
$$\underbrace{\hspace{5em}}_{\underline{\kappa}^2 \underline{q}^2 + \underline{\mathcal{I}}_3^2 \underline{e}^{3+} := \underline{e}^{2+}}$$

$$\underbrace{\hspace{5em}}_{\underline{\kappa}^1 \underline{q}^1 + \underline{\mathcal{I}}_2^1 \underline{e}^{2+} := \underline{e}^{1+}}$$

$$U^{1+} \approx \frac{1}{2}(\underline{q}^1)^\top \underline{e}^{1+} = \frac{1}{2} \sum_{\mathbf{m}} (q^1)_{\mathbf{m}} (e^{1+})_{\mathbf{m}} = \frac{1}{2} \sum_i q_i \underbrace{\sum_{\mathbf{m}} (e^{1+})_{\mathbf{m}} \varphi_{\mathbf{m}}^1(\mathbf{r}_i)}_{\text{electrostatic potential at } \mathbf{r}_i}$$

More algorithm details: “Ladder diagram”

The “ladder diagram” is nothing more than a pictorial representation of the formula expressions from the previous slide.



from: J. Chem. Phys. 144, 114112 (2016)

More algorithm details: Calculation of forces

So far, only calculation of energies:

$$U \approx \underbrace{\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} q_i q_j k_0(|\mathbf{r}_i - \mathbf{r}_j|) - \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N q_i^2 k_l(r) \big|_{r=0}}_{U^0} + \underbrace{\frac{1}{2} \sum_i q_i \sum_{\mathbf{m}} (e^{1+})_{\mathbf{m}} \varphi_{\mathbf{m}}^1(\mathbf{r}_i)}_{U^{1+}}$$

Force on the i -th particle:

$$\mathbf{F}_i = -\nabla_i U$$

- Requires the gradients $\nabla k_0(|\mathbf{r}|)$ and $\nabla \varphi_{\mathbf{m}}^1(\mathbf{r}) \rightarrow$ analytic expressions available.
- The “grid representation of the electrostatic potential” $\underline{e}^{1+}$, calculated during energy evaluation, can be reused in force calculation.

Schematic cost analysis

Schematic cost analysis

$$U = U^0 + U^{1+} \approx \underbrace{\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} q_i q_j k_0(r_{ij}) - \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N q_i^2 k_l(r)|_{r=0}}_{U^0} + \underbrace{\frac{1}{2} \sum_{l=1}^L \sum_{\mathbf{m}} \sum_{\mathbf{n}} q_{\mathbf{m}}^l \mathcal{K}_{\mathbf{m},\mathbf{n}}^l q_{\mathbf{n}}^l}_{\approx U^{1+}}$$

Scaling of U^0 with number of particles N :

- First term scales as $\mathcal{O}(N)$ because of the short range of $k_0(r)$.
- Second term does not even contain a sum over pairs of particles.

Schematic cost analysis

$$U = U^0 + U^{1+} \approx \underbrace{\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} q_i q_j k_0(r_{ij}) - \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N q_i^2 k_l(r) \Big|_{r=0}}_{U^0} + \underbrace{\frac{1}{2} \sum_{l=1}^L \sum_{\mathbf{m}} \sum_{\mathbf{n}} q_{\mathbf{m}}^l \mathcal{K}_{\mathbf{m},\mathbf{n}}^l q_{\mathbf{n}}^l}_{\approx U^{1+}}$$

Scaling of U^{1+} with number of particles N :

- Computation of “grid charges” $q_{\mathbf{m}}^l = \sum_{i=1}^N q_i \varphi_{\mathbf{m}}^l(\mathbf{r}_i)$ is linear in N
- Cost of evaluating interactions determined by number of non-zeros in the $\mathcal{K}_{\mathbf{m},\mathbf{n}}^l$
 - Grid level $l = L$: infinite-range interaction kernel
(# non-zeros) = (# grid points)²
 $\Rightarrow$ choose L such that number of grid points $\leq \sqrt{N}$
 - Grid levels $l = 1, \dots, L - 1$: finite-range interaction kernels
(# non-zeros) $\approx \left(\frac{\text{cutoff at level } l}{\text{grid spacing at level } l} \right)^3 = \text{const.}$

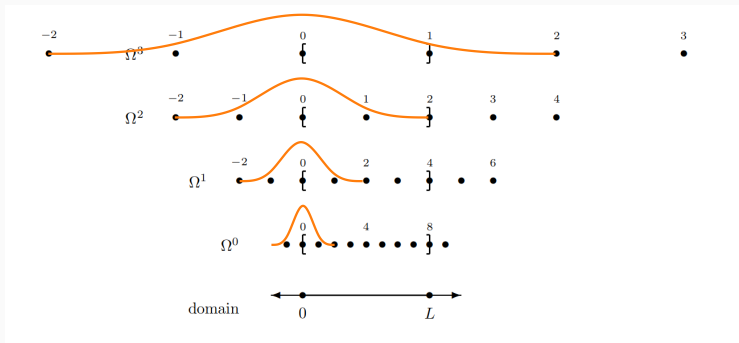
Boundary conditions

This part is based on:

Multilevel Summation for the Fast Evaluation of Forces for the Simulation of Biomolecules., David J. Hardy. Ph.D. thesis, University of Illinois at Urbana-Champaign, 2006.

Boundary conditions: Non-periodic

Need to extend grid boundary to avoid information loss in interpolation.



adapted from: David J. Hardy. Ph.D. thesis, 2006.

Periodic boundary conditions: General

Electrostatic energy *per unit cell* in a periodic system:

$$U = \underbrace{\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} \frac{q_i q_j}{|\mathbf{r}_i - \mathbf{r}_j|}}_{\text{interaction within unit cell}} + \underbrace{\frac{1}{2} \sum_{\mathbf{T} \neq \mathbf{0}} \sum_{i=1}^N \sum_{j=1}^N \frac{q_i q_j}{|\mathbf{r}_i - \mathbf{r}_j + \mathbf{T}|}}_{\text{interaction with periodic copies}}, \quad \text{where}$$

- i, j run over particles inside one unit cell,
- $\mathbf{T}$ are all possible translation vectors between unit cells.

Note:

- Diverges for charged systems, i.e., whenever $\sum_{i=1}^N q_i \neq 0$.
- Even for charge-neutral systems, only conditionally convergent, i.e., value of the sum depends on order of terms.

MSM for periodic boundary conditions: formal

With the familiar splitting $1/r = \sum_{l=0}^L k_l(r)$, the electrostatic energy becomes

$$\begin{aligned}
 U &= \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N q_i q_j k_0(|\mathbf{r}_i - \mathbf{r}_j|) + \frac{1}{2} \sum_{\mathbf{T} \neq \mathbf{0}} \sum_{i=1}^N \sum_{j=1}^N q_i q_j k_0(|\mathbf{r}_i - \mathbf{r}_j + \mathbf{T}|) \\
 &\quad + \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j \neq i}^N q_i q_j k_l(|\mathbf{r}_i - \mathbf{r}_j|) + \frac{1}{2} \sum_{l=1}^L \sum_{\mathbf{T} \neq \mathbf{0}} \sum_{i=1}^N \sum_{j=1}^N q_i q_j k_l(|\mathbf{r}_i - \mathbf{r}_j + \mathbf{T}|) \\
 &= \underbrace{\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N q_i q_j k_0(|\mathbf{r}_i - \mathbf{r}_j|) + \frac{1}{2} \sum_{\mathbf{T} \neq \mathbf{0}} \sum_{i=1}^N \sum_{j=1}^N q_i q_j k_0(|\mathbf{r}_i - \mathbf{r}_j + \mathbf{T}|)}_{:= U^0} - \frac{1}{2} \sum_{i=1}^N q_i^2 k_l(r)|_{r=0} \\
 &\quad + \underbrace{\frac{1}{2} \sum_{l=1}^L \sum_{\mathbf{T}} \sum_{i=1}^N \sum_{j=1}^N q_i q_j k_l(|\mathbf{r}_i - \mathbf{r}_j + \mathbf{T}|)}_{:= U^{1+}}
 \end{aligned}$$

MSM for periodic boundary conditions: formal

Looking at U^{1+} , we can identify new effective interaction kernels g_l :

$$U^{1+} = \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N q_i q_j \underbrace{\sum_{\mathbf{T}} k_l(|\mathbf{r}_i - \mathbf{r}_j + \mathbf{T}|)}_{:=g_l(\mathbf{r}_i, \mathbf{r}_j)} = \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N q_i q_j g_l(\mathbf{r}_i, \mathbf{r}_j)$$

→ Looks formally identical to non-periodic version!

MSM for periodic boundary conditions: formal

Interpolation similarly to non-periodic case:

$$U^{1+} = \frac{1}{2} \sum_{l=1}^L \sum_{i=1}^N \sum_{j=1}^N q_i q_j g_l(\mathbf{r}_i, \mathbf{r}_j) \approx \frac{1}{2} \sum_{l=1}^L \sum_{\mathbf{m}} \sum_{\mathbf{n}} q_{\mathbf{m}}^l \mathcal{G}_{\mathbf{m},\mathbf{n}}^l q_{\mathbf{n}}^l$$

Now, in periodic case:

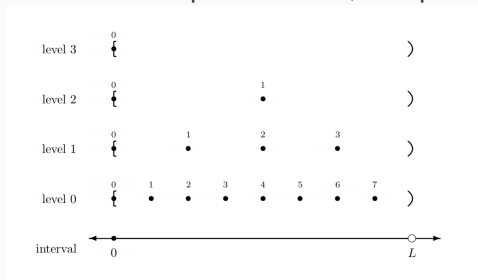
Keep going to higher grid levels until highest level L has only *one* point:

- The grid charges have the property $\sum_{\mathbf{m}} q_{\mathbf{m}}^l = \sum_{i=1}^N q_i$.
- Single grid point and charge-neutrality ($q_{\mathbf{m}_1}^L = 0$)
 $\implies$ No contribution from highest grid level L !
 - We do not need to treat the problematic infinite-range kernel k_l .
 - Truncation error turned into interpolation error.

MSM for periodic boundary conditions: in practice

In practice, periodic boundary conditions work almost like non-periodic ones, except

- No extension of grids beyond simulation cell boundary.
- All operations performed on grid charges $\underline{q}$ and potentials $\underline{e}^{I+}$ ($\underline{\mathcal{I}}$ and $\underline{\mathcal{K}}$) wrap around through grid boundaries.



from: David J. Hardy. Ph.D. thesis, 2006.

End of presentation